

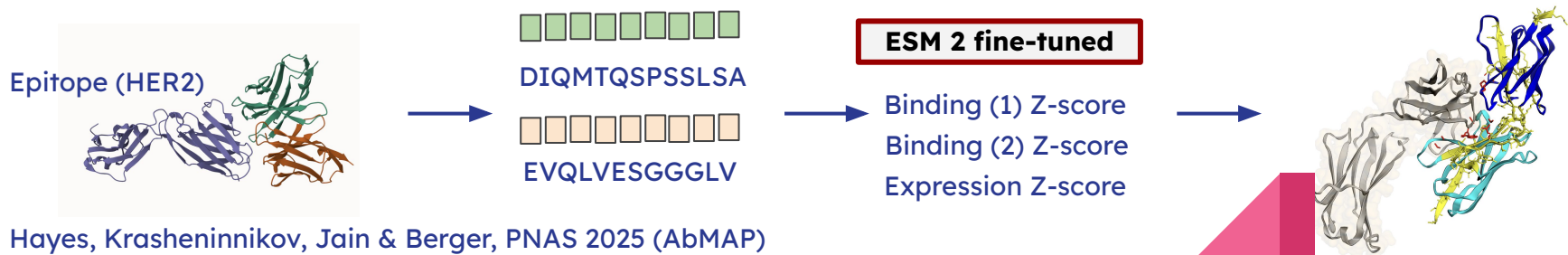


Redesigning Humanized Monoclonal Antibodies Through Computational Approaches

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Motivation

- Development cost of therapeutic antibodies (Ab) : **8-9 years** and **650 million USD**
- **Binding** and **Developability** are essential tradeoffs in traditional Ab pipeline
- Following **AbMAP**, which adapts foundational protein language models for antibody-specific representation learning
- Our **Germinal**-inspired pipeline: RFdiffusion/RFantibody backbone generation → ProteinMPNN sequence design → finetuned ESM-2 candidate scoring → Chai-1 structural validation.



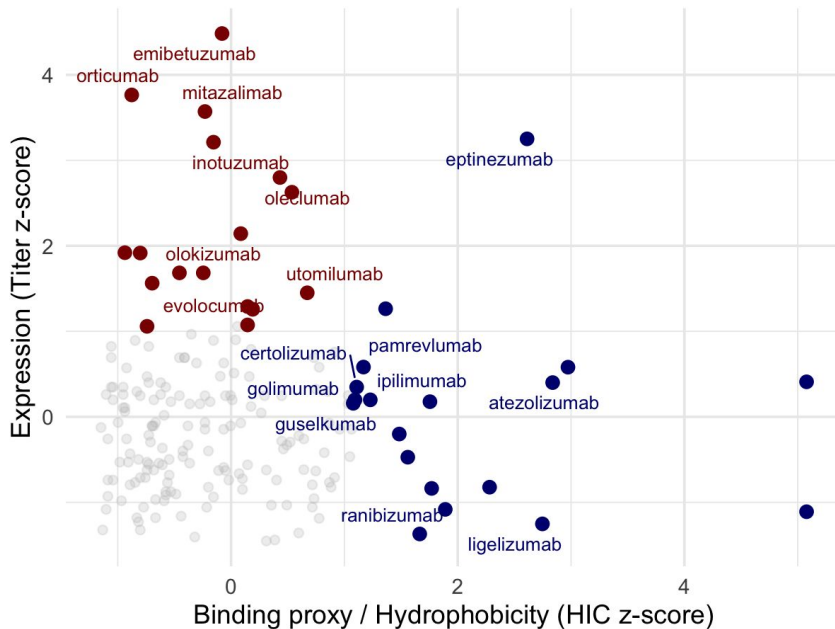
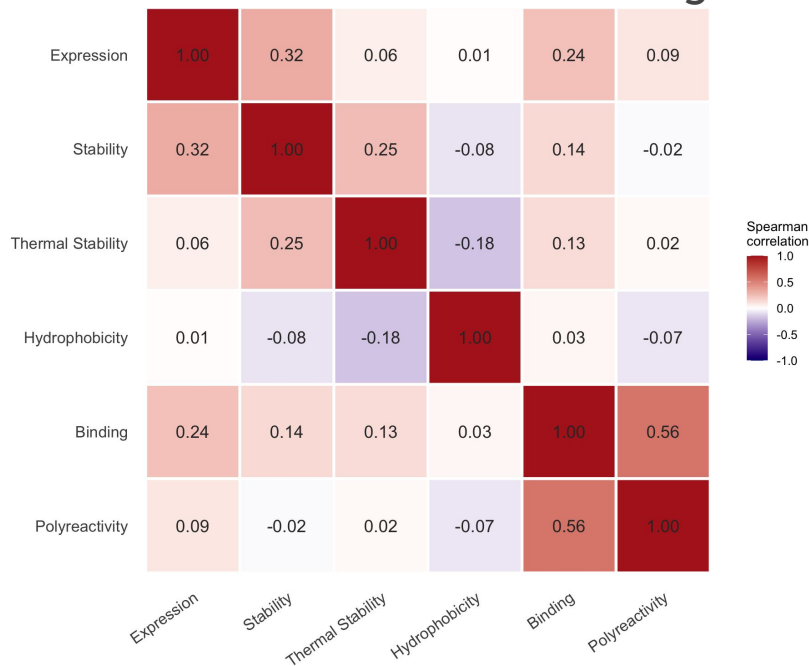
Hayes, Krasheninnikov, Jain & Berger, PNAS 2025 (AbMAP)

Mille-Fragoso, Driscoll, Wang, Dai, Widatalla, Zhang, Zhang, Rao, Feng, Hie & Gao, bioRxiv 2025 (Germinal)

<https://naturalantibody.com/use-case/business-benefits-of-ai-driven-antibody-drug-discovery/>

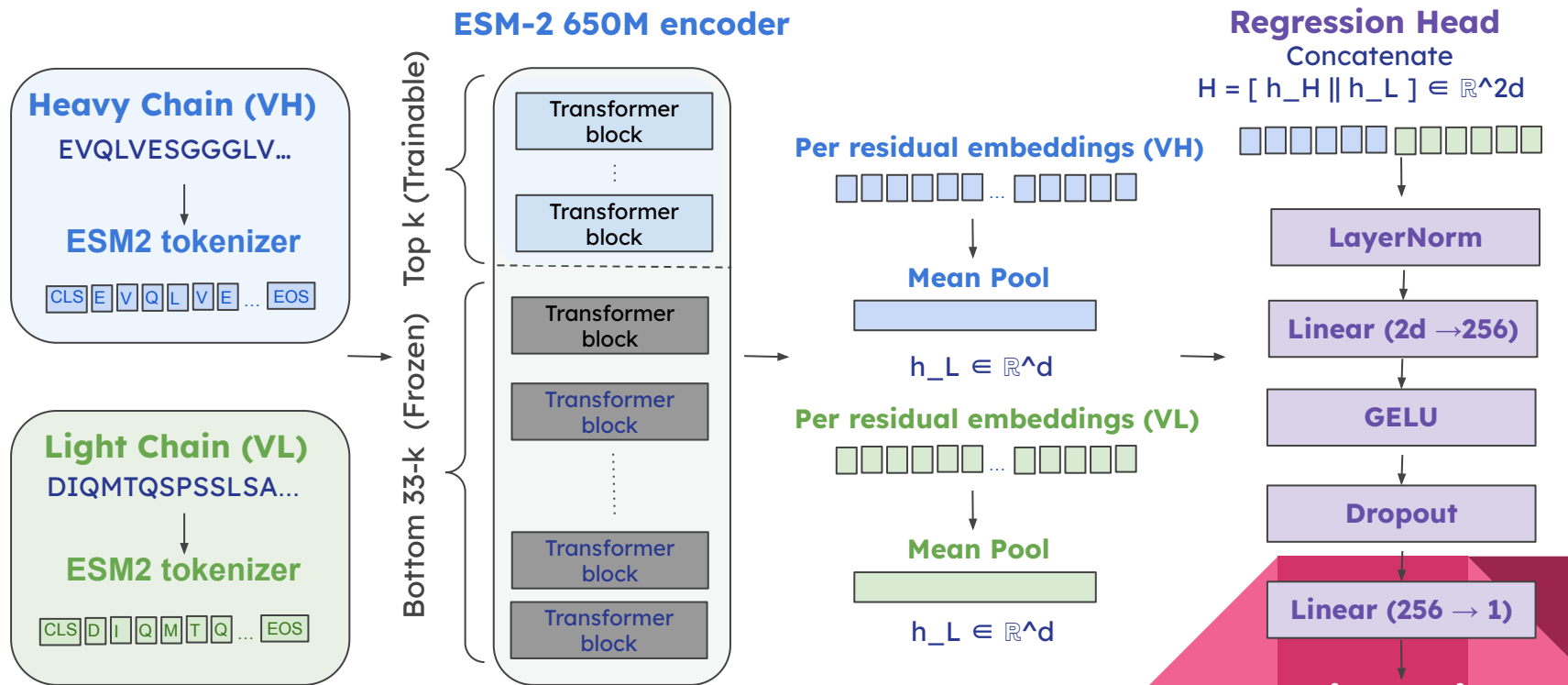
Hypothesis

- **Binding** and **Expression** are essential tradeoffs in traditional Ab pipeline
- We **hypothesize** that **multi-objective scoring** for antibody designs facilitates optimization selection



Data analyzed from van Niekerk et al., *mAbs* 2026 (Ginkgo Datapoints Antibody Developability Competition)

Scoring/Surrogate Model Architecture



Loss function : MSE of predicted and experiment antibody fitness

Training Dataset

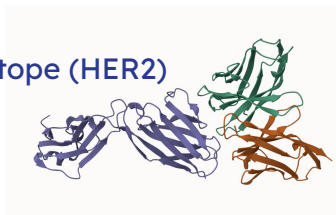
- Publicly available FLAb : Fitness Landscapes for Antibodies database
 - **Koenig** binding and expression subset : 4,275 single-mutant antibody variants from anti-VEGF antibody
 - **Warszawski** binding subset : 2,048 single-mutant antibody variants from anti-lysozyme antibody
- Each dataset were treated as separate tasks based of different experimental methods, held out with 10-fold cross validation.

Antibody Generating Pipeline

RFdiffusion/ RFantibody

Diffusion-based
antibody backbone
generation
conditioned on CDR

Epitope (HER2)



ProteinMPNN

Graph based encoder
(structure) - decoder
(sequence)
framework

DIQMTQSPSSL
EVQLVESGGGLV

ESM2 scoring

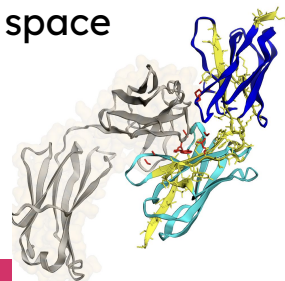
ESM2 based Fitness
prediction from 3
tasks yielded a
composite score for
reranking

ESM 2 fine-tuned

Binding (1) Z-score
Binding (2) Z-score
Expression Z-score

Chai-1 evaluation

Transformer networks
and diffusion-based
structure refinement
to predict binding in
3D space



Complementarity-determining regions (CDRs) ; HER2 (human epidermal growth factor receptor 2)

Phase 0: Baseline Fitness Prediction

Spearman ρ (10-fold cross validation)

Flab Benchmarks

(3 tasks)

- Koenig Binding
- Koenig Expression
- Warszawski Binding

Method	Koenig Bind.	Koenig Expr.	Wars. Bind.
Mean baseline	0.000	0.000	0.000
Sequence + ridge regression	0.174	0.247	0.260
AbMAP + ridge regression	0.334	0.544	0.345
ESM-2 + MLP	0.389	0.727	0.390
ESM-2 + ridge regression	0.466	0.729	0.479

Phase 1: Fine-tuning ESM-2 (650M)

Does task-specific fine-tuning meaningfully improve the frozen ESM-2 surrogate?

- Sweep on **ESM-2 35M** backbone $lr \in \{1e-4, 1e-5, 1e-6\}$, unfrozen layers $d \in \{0, 2, 4\}$
- Best config **transferred** (650M): $lr = 1e-4$, $d = 4$

Task	Frozen ρ	Fine-tuned ρ	$\Delta\rho$
Koenig Bind.	0.466	0.522	+0.056
Koenig Expr.	0.729	0.809	+0.080
Warszawski Bind.	0.479	0.555	+0.076

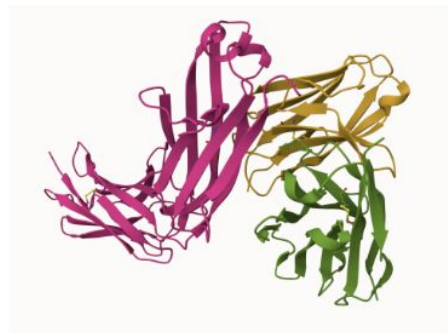
Finding: Consistent +0.057—0.080 Spearman gain across all 3 tasks.
Expression task benefits most.

Phase 2: HER2 Candidate Generation Workflow

- **Target:** HER2-a cancer-related cell-surface receptor protein.
- **Input:** HER2-trastuzumab complex (PDB: 1N8Z) and scFv scaffold
- **Model:** RFdiffusion + ProteinMPNN
- **Output:** 40 candidates passed to Phase 3 for ESM-2 reranking



(a) Germinal trastuzumab–HER2 (HER2 green, trastuzumab orange).



(b) scFv scaffold–HER2 (HER2 magenta, VH green, VL yellow).

Figure 1. Phase 2 generative antibody structures visualized in Mol*. Left: Germinal-generated trastuzumab scFv–HER2 complex using the 1N8Z template. Right: scFv antibody scaffold used for RFantibody case study docking to HER2 via PyDockWEB.

Phase 3: Surrogate Scoring & Pareto Reranking

Among 40 generated candidates, which best balance predicted binding and expression?

- Each candidate scored by fine-tuned ESM-2 on binding z-score (both Koenig and Warszawski) and expression z-score

Pareto front selection: preserve tradeoff between binding/yield trade off

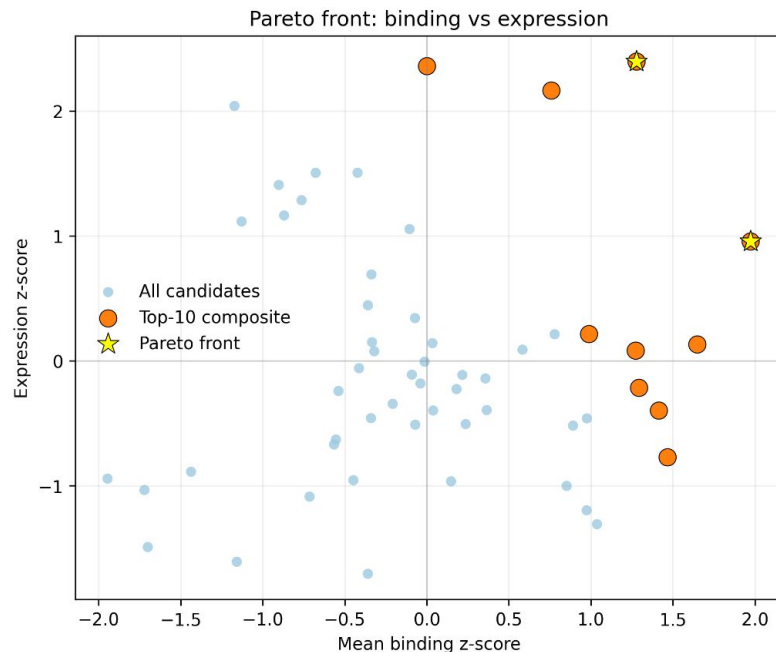


Figure 2. HER2 Pareto front. Each blue point is a generated antibody candidate scored by the fine-tuned ESM-2 surrogate. Yellow star indicates the Pareto-optimal candidates, each maximizing either mean binding and expression z-score without being dominated on the other objective.

Phase 3: Surrogate Scoring & Pareto Reranking

Pareto-ranked candidates show balance between binding and expression.

Rank	Candidate ID	Binding 1 (z-score)	Binding 2 (z-score)	Expression (z-score)	Composite Score
1	1N8Z_gen4_seq4	1.783	1.622	1.298	1.568
2	1N8Z_gen2_seq9	1.559	1.244	0.929	1.244
3	1N8Z_gen3_seq8	1.369	1.210	1.050	1.210
4	1N8Z_gen5_seq3	1.192	0.952	0.712	0.952
5	1N8Z_gen2_seq7	0.962	0.991	1.021	0.991

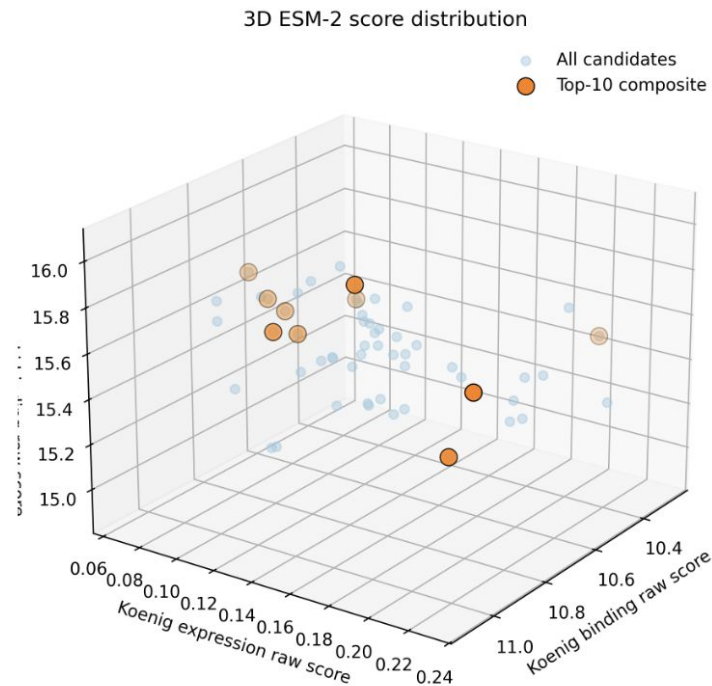


Figure 3. HER2 3D raw score. Each point represents Koenig binding, Koenig expression, and Warszawski binding raw scores of the candidates. Orange dots highlights the top candidates from the ESM2 finetuned model scoring

Phase 4: Chai-1 Structural Validation

	ESM-2	Random Baseline
ipTM	0.610	0.601
pTM	0.733	0.722

Δ ipTM = +0.009, Δ pTM = +0.011 (n = 5 per group, Welch's t-test $p > 0.05$)

- ESM-2 reranked candidates show consistently higher structural confidence
- Small n limits statistical significance

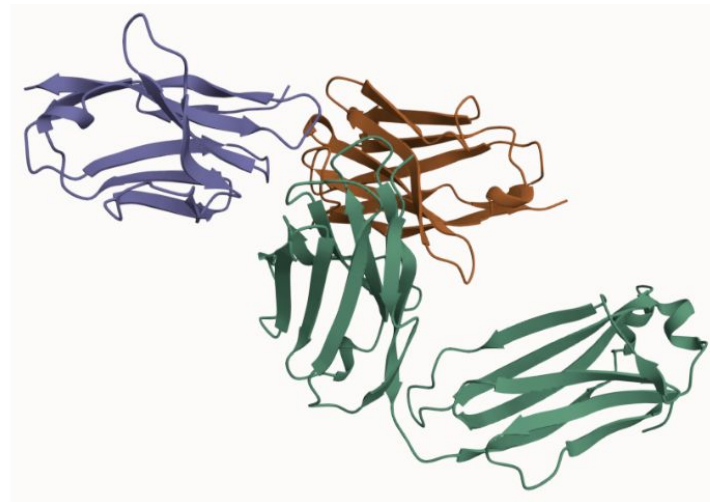


Figure 5. Chai-1 predicted structure of a top ESM-2-reranked HER2 antibody candidate (pred.model_idx_0). Green: HER2 target (Entity A); orange: VH chain (Entity B); purple: VL chain (Entity C). The antibody Fv region docks against the HER2 extracellular domain, consistent with the elevated ipTM scores of ESM-2-selected candidates.

Conclusions & Future Work

Our contribution: A modular, end-to-end pipeline combining fine-tuned ESM-2 surrogate scoring with structured-based generative design, reducing reliance on wet-lab evaluation for early-stage antibody candidate prioritization.

Key takeaways

- Fine-tuning on ESM-2 yields consistent $\Delta p = +0.056 - 0.080$ across all 3 FLAb tasks
- Pareto reranking selects candidates with higher structural confidence

Future Work

- Structure-conditioned scores eg: ESMFold would improve reranking fidelity
- Pipeline scales to additional therapeutic targets (PD-L1, EGFR)

Thank you!